



OS14 - Identifying Threats to Technical Value Delivery

Team 14A: David Ricardo, Camden Ramsay, Nora Alshareef, Michel Fernandes
11 March 2026

Our current perspective is that Sanguine is best understood as a physician-facing clinical interpretation layer for laboratory results, rather than as a general medical chatbot or a simple reporting add-on. The system aims to improve Mayo Clinic lab patient care outcomes at scale by augmenting physician-facing test results with relevant research through an agentic AI clinical librarian. Our prior work in OS11 and OS13 clarified the system boundary, identified the most important architectural decisions, and showed that concepts closer to the Mayo Model of Care can create value, but with materially different cost and governance implications. In OS14, our perspective is focused on the uncertainty surrounding our design decisions and concepts.

Q1: Characterizing Uncertainty

We created probability density functions (PDFs) for Clinical Accuracy (M6) and dollars per result (M8) for every option included in the tradespace (D2, D3, D5, D6, D9, and D11). For M8, we used cost per interpreted result rather than total annual cost because the goal of OS14 is to capture uncertainty from architectural choices. We calculated this value by dividing the annual cost estimates by the yearly laboratory volume (from OS13 data). This normalization makes concepts directly comparable and better isolates the architecture's cost impact. Thus, the Pareto frontier, shown in Q3 of OS13, also had to be recalculated. Even so, it still shows the same behavior seen in OS13.

		Performance metric: Clinical Accuracy (M6)			Cost metric: Dollars per result (M8)		
Decision	Option	PDF	Params	Rationale	PDF	Params	Rationale
D2: Model Origin Sponsor Input: Provider choice affects operational stability and governance constraints.	Gemini Family Only	Beta	$\alpha=21$ $\beta=9$ $\mu=0.7$ $k=30$	Lower variability because of a single provider reduces model dispersion.	Log-Normal	$\mu=0.038$, CV=0.3	Moderate variability because pricing is predictable, but token usage varies with context.
	Other closed source	Beta	$\alpha=15$ $\beta=10$ $\mu=0.6$ $k=25$	Higher uncertainty due to multiple providers' models and configurations.	Log-Normal	$\mu=0.077$, CV=0.4	Higher variability due to pricing uncertainty and integration overhead.
	Train our own	Beta	$\alpha=12$ $\beta=3$ $\mu=0.8$ $k=15$	Higher uncertainty due to dependency on data coverage and training.	Log-Normal	$\mu=0.462$, CV=0.5	High variability due to periodic training and evaluation costs.
	Open source (self-hosted)	Beta	$\alpha=10$ $\beta=10$ $\mu=0.5$ $k=20$	Higher uncertainty due to dispersion of models and clinical alignment.	Log-Normal	$\mu=0.154$, CV=0.5	High variability due to utilization and hosting costs.
D3: Research Paradigm Sponsor Input: Not provided for this specific decision, but scheduled for future discussions.	RAG	Beta	$\alpha=17.5$ $\beta=7.5$ $\mu=0.7$ $k=20$	Variable due to dependency on corpus quality.	Log-Normal	$\mu=0.077$, CV=0.35	Moderate variability due to additional retrieval costs.
	Fine-tuning	Beta	$\alpha=14$ $\beta=6$ $\mu=0.7$ $k=20$	Variable due to drift risk and data dependency.	Log-Normal	$\mu=0.115$, CV=0.4	High variability due to retraining and evaluation costs.
	RAG + Fine-tuning	Beta	$\alpha=14.4$ $\beta=3.6$ $\mu=0.8$ $k=18$	Variable because of retrieval and drift risk.	Log-Normal	$\mu=0.192$, CV=0.45	High variability due to combined retrieval, retraining, and evaluation costs.
	Rely on Pre-training	Beta	$\alpha=9$ $\beta=21$ $\mu=0.3$ $k=30$	More consistent but with lower accuracy because there is no domain adaptation.	Deterministic	$v=0$	Zero incremental cost beyond baseline inference because there is no added clinical retrieval or training.
D5: Number of AI roles per test result	Non-agentic AI	Beta	$\alpha=3.5$ $\beta=31.5$ $\mu=0.1$ $k=35$	Lower accuracy and lower variability because there is no role decomposition.	Deterministic	$v=0$	Zero incremental role cost because there are no additional agents.
	Single agent	Beta	$\alpha=12.5$ $\beta=12.5$ $\mu=0.5$ $k=25$	Moderate variability due to case complexity.	Log-Normal	$\mu=0.038$, CV=0.35	Moderate variability in longer contexts of rare cases.
	Small group	Beta	$\alpha=14$ $\beta=3.6$ $\mu=0.8$ $k=18$	Higher accuracy with variation due to cross-validation.	Log-Normal	$\mu=0.115$, CV=0.45	High variability due to the coordination of multiple agents.

		Performance metric: Clinical Accuracy (M6)			Cost metric: Dollars per result (M8)		
Decision	Option	PDF	Params	Rationale	PDF	Params	Rationale
	Large group	Beta	$\alpha=10.8$ $\beta=1.2$ $\mu=0.9$ $k=12$	Higher accuracy with more variability with coordination effects.	Log-Normal	$\mu=0.2569$, CV=0.55	High variability due to the coordination of multiple agents.
D6: Data sources Sponsor Input: Coverage gaps appear primarily in rare tests.	Mayo internal only	Beta	$\alpha=13.2$ $\beta=8.8$ $\mu=0.6$ $k=22$	Moderate variability due to coverage in rare cases.	Deterministic	$v=0$	Zero incremental cost due to internal sources with no licensing (sunk cost).
	Mayo + single external source	Beta	$\alpha=14.4$ $\beta=3.6$ $\mu=0.8$ $k=18$	Variable due to harmonization and consistency of sources.	Log-Normal	$\mu=0.058$, CV=0.4	High variability due to external access and latency.
	Multiple internal + external sources	Beta	$\alpha=13.5$ $\beta=1.5$ $\mu=0.9$ $k=15$	More variables due to conflicts and duplication between references.	Log-Normal	$\mu=0.154$, CV=0.5	High variability due to multi-source retrieval overhead.
	Open Internet only	Beta	$\alpha=5.4$ $\beta=12.6$ $\mu=0.3$ $k=25$	Higher uncertainty due to limitations in source quality.	Log-Normal	$\mu=0.077$, CV=0.60	High variability due to the search algorithm, filtering, and source review.
D9: Clinical validation and release control Sponsor Input: Human review is conditional (40% common, 60% rare).	Rule-based condition	Beta	$\alpha=15$ $\beta=15$ $\mu=0.5$ $k=30$	Moderate variability due to not covering all interpretation errors.	Triangular	$a=0.01$ $c=0.019$ $b=0.038$	Low variability in operational efforts.
	Tiered release	Beta	$\alpha=16$ $\beta=4$ $\mu=0.8$ $k=20$	Variable due to threshold and routing sensitivity.	Triangular	$a=0.2$ $c=0.577$ $b=1.2$	Higher variation in escalated cases efforts.
	Human-in-the-loop	Beta	$\alpha=28.5$ $\beta=1.5$ $\mu=0.95$ $k=30$	High accuracy with moderate variability due to Human review.	Triangular	$a=4$ $c=7.692$ $b=15$	Long-tail costs variability due to throughput and conditional escalations with gate review.
D11: Model specification Sponsor Input: Not provided for this specific decision, but scheduled for future discussions.	Efficient, Text-only	Beta	$\alpha=7.5$ $\beta=17.5$ $\mu=0.3$ $k=25$	Lower accuracy in complex cases, with variation depending on case complexity.	Log-Normal	$\mu=0.038$, CV=0.3	Low variation due to predictable token usage.
	Reasoning, Medium price	Beta	$\alpha=14$ $\beta=6$ $\mu=0.7$ $k=20$	Higher accuracy with variation due to context length.	Log-Normal	$\mu=0.154$, CV=0.4	Moderate variability due to context and retries.
	Multimodal, powerful, expensive	Beta	$\alpha=14.4$ $\beta=3.6$ $\mu=0.8$ $k=18$	Higher accuracy, but with greater variation due to case complexity.	Log-Normal	$\mu=0.385$, CV=0.45	High variability due to large context calls.
	Mix multiple models	Beta	$\alpha=12$ $\beta=4$ $\mu=0.8$ $k=15$	Variable due to routing and boundary cases.	Log-Normal	$\mu=0.115$, CV=0.5	High variability due to model routing across different pricing tiers.

We modeled **M6 (accuracy)** with a **Beta distribution** because it is a proportion limited to the range of 0 to 1, and this family is flexible to represent both high uncertainty and high confidence. In practice, we define the mean as our prior expectation (from mapping the OS13 accuracy score) and use $\alpha+\beta$ to encode prior confidence (a larger k implies lower variance). This choice was validated with our project sponsor, Dr. Chris Garcia from Mayo Clinic, who confirmed M6 as a good proxy for clinical performance. We modeled **M8 (cost)** using **Log-Normal and Triangular distribution** because costs are always positive, and often right-skewed. We parameterized this distribution with the cost per result from OS13 by setting the mean to the annual cost divided by yearly volume and an operational variability (coefficient of variation, CV) that reflects operational variability. In decisions involving operational policy or human workflow, we used a Triangular distribution instead because the stakeholders can provide specific bounds or most likely values. Dr. Garcia also mentioned that human review is a conditional policy (40% of common tests and 60% of rare tests), rather than a fixed mode applied to every result. We evaluated this paradigm of review and determined it aligns well with Sanguine, so we treated human review with a gated cost model: zero when is not triggered and triangular when it is. Finally, for cases where it is effectively zero, we represented it as deterministic.

This is a simple model, aligned with [Piazza @84](#) and option B from [Piazza @94](#). This simplification is valid because all the architectural decisions contribute directly to “intelligence”, which is subjective and directly determines accuracy. To scale this up, we’d want to create a more detailed model with internal parameters. But for now, this model is simple and insightful.

Q2: Selection of Architectures

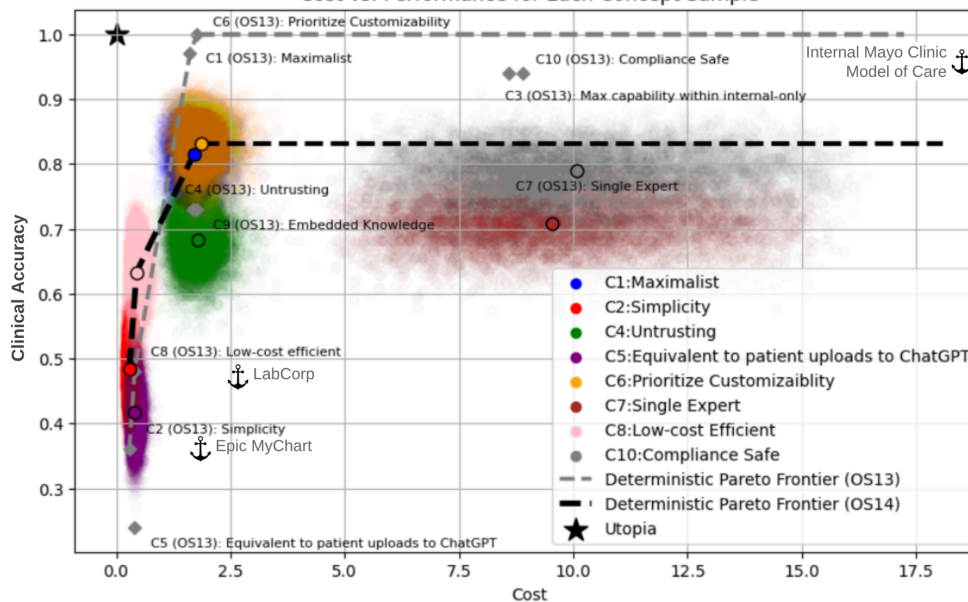
For an uncertainty analysis, we selected 8 concepts covering different regions of the OS13 tradespace, rather than concentrating only on the deterministic points. The selection preserves variety in the most relevant architectural decisions and includes the team's preferred concept, a low-complexity baseline, concepts on the Pareto frontier, and dominated concepts used as contrast cases. This allows us to assess not only the average concept performance of but also the uncertainty surrounding architectures with different implications of options.

ID	Name	D2: Model Origin	D3: Research Paradigm	D5: Number of AI roles per test result	D6: Data sources	D9: Release control	D11: Model Specification	Role	Rationale
C1	Maximalist	Gemini family only	RAG + Fine-tuning	Large group (5 - 10)	Multiple internal + external sources	Tiered release	Multimodal, powerful, expensive (3 Pro)	Preferred	Included to represent the high-performance end of the tradespace and the team's preferred architecture.
C2	Simplicity	Gemini family only	Rely on pre-training	1 (single agent)	Mayo internal only	Rule-based condition	Efficient, Text-only (Flash)	Baseline	Included to provide a low-complexity baseline for comparison across all uncertainty scenarios.
C4	Untrusting	Train our own	Rely on pre-training	Small group (2-5)	Mayo internal only	Tiered release	Multimodal, powerful, expensive (3 Pro)	Frontier	Included to represent a provider-independence strategy with stronger internal control.
C5	Equivalent to patient uploads to ChatGPT	Other closed source	Rely on pre-training	1 (single agent)	Open Internet only	Rule-based condition	Efficient, Text-only (Flash)	Dominated	Included as a dominated reference case with weak grounding and limited compliance safeguards.
C6	Prioritize Customizability	Train our own	RAG + Fine-tuning	Large group (5-10)	Multiple internal + external sources	Tiered release	Mix multiple models	Frontier	Included to capture the most customizable concept.
C7	Single Expert	Other closed source	RAG + Fine-tuning	1 (single agent)	Mayo internal only	Human-in-the-loop	Multimodal, powerful, expensive (3 Pro)	Dominated	Included to isolate the uncertainty effects of concentrating reasoning in a single-agent architecture.
C8	Low-cost Efficient	Other closed source	RAG + Fine-tuning	1 (single agent)	Mayo internal only	Rule-based condition	Mix multiple models	Frontier	Included to represent a lower-cost frontier concept with competitive performance.
C10	Compliance Safe	Train our own	RAG Only	Large group (5-10)	Mayo internal only	Human-in-the-loop	Multimodal, powerful, expensive (3 Pro)	Dominated	Included as a dominated control-heavy concept to test whether additional assurance justifies the added cost.

For the Monte Carlo analysis, we implemented a [Python](#) simulation based on the probability distributions defined in Q1 for cost (M8) and clinical performance (M6) in each architectural option. Each concept was modeled as a fixed combination of the six architectural decisions, and, at each iteration, the script randomly generated a cost value and a performance value for each decision within the concept. The sampled costs were summed to represent the total cost per interpreted outcome, while performance was estimated by the simple average of the sampled performance values across those same decisions. This simplification assumes that each of the selected options contributes equally to the final estimate of clinical accuracy. This process was repeated 10,000 times per concept, generating an empirical distribution of possible outcomes for each concept. Note that we removed concept C3 from OS13 because it was neither insightful nor close to being the best choice.

We show the uncertainty in two tradespace visualizations built from the same simulation results. The first shows the point cloud for each concept and highlights each concept's mean along both axes. We draw a deterministic pareto front over these mean

Cost vs. Performance for Each Concept Sample



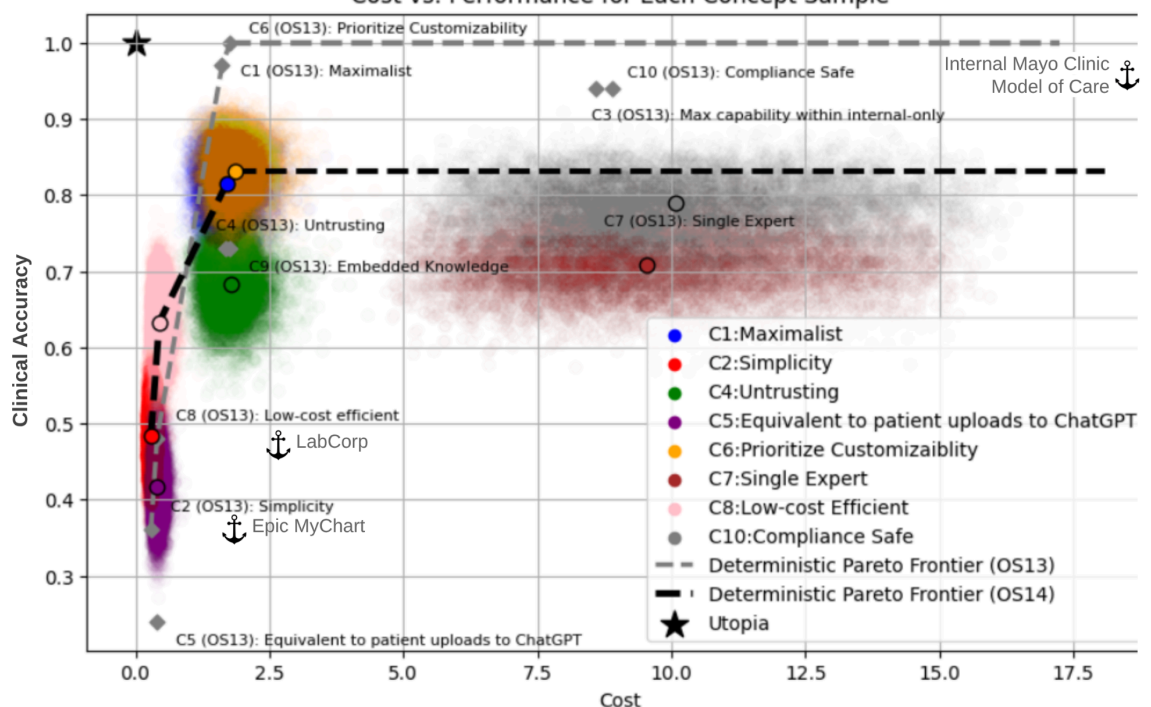
values. We also include the deterministic pareto front from OS13 in grey for context, though the pareto front has moved as our model has improved. We show previous concepts found in OS13 Q2 with the anchor icon (⚓). The internal mayo clinic model of care is shown on the chart for context, but it's far more expensive than everything else. It should properly be at least an order of magnitude to the right, but that would warp the chart.

The second tradespace highlights the uncertainty region for each concept and the fuzzy Pareto frontier, allowing us to compare not only the average positions of the alternatives but also the extent and orientation of their variation. Since this is deeper into OS14 and the second chart is already busy, OS13 content is removed.

In the first tradespace, the main comparison was between the deterministic reading of OS13 and the simulated distribution of concepts under uncertainty. This visualization showed that the main change in interpretation occurs in the concepts more focused on compliance and governance. In OS13, concepts such as C7 ("Single Expert") and C10 ("Compliance Safe") seemed strong because they incorporated more rigorous validation and human-in-the-loop review, despite their high cost. After the Monte Carlo simulation, however, C1 ("Maximalist") and C6 ("Prioritize Customizability") were positioned more favorably relative to the Pareto frontier, because they maintained high performance without the same increase in cost and variability seen in the more rigorous clinical control concepts. Thus, the tradespace made it clearer that additional validation controls don't necessarily preserve the same relative advantage once uncertainty is incorporated into the analysis.

In the second tradespace, the emphasis was on the explicit representation of uncertainty around each concept. In addition to the mean position of each alternative, we used confidence ellipses constructed from the joint distribution of cost and performance samples. The ellipses were chosen because they show not only the dispersion on each axis but also the orientation of the variability in two-dimensional space, making the relationship between cost and performance for each concept clearer. We used the 68% and 95% levels because they are standard statistical reference levels to represent, respectively, the most typical region of the results and a wider range of plausible results. Thus, the second tradespace augments the first by showing not only where each concept tends to be located, but also how much its position can vary under uncertainty.

Cost vs. Performance for Each Concept Sample



MEMO

To: Dr. Chris Garcia, Mayo Clinic
From: Team 14A
Date: March 2026
Subject: Impact of Architectural Uncertainty on
Clinical AI Value Delivery

Executive Summary

Our Monte Carlo analysis of eight architectures shows that uncertainty changes how the OS13 tradespace should be interpreted. Most of the variation in clinical accuracy and cost per interpreted result comes from just a few architectural decisions. At this stage, architectures that combine adaptability with appropriate validation seem to hold up better than those built around intensive human review.

Major Sources of Uncertainty and Sensitivity of Valued Outcomes:

- **Clinical Accuracy** (M6) is highly sensitive to metrics D5, D6, and D9.
Using **multiple agent roles** (D5) can provide more accurate expectations, but it also adds coordination overhead and makes interactions less predictable as the number of roles increases. Pulling from multiple **data sources** (D6) expands coverage, but it also increases data harmonization issues and retrieval inconsistency. Within our framework of **clinical validation and release control** (D9), human in the loop review can improve accuracy, but it also can further introduce variability within outcomes, review escalation determination can hinge on factors such as case complexity and rare conditions.
- **Cost per Result** (M8) is highly sensitive to metrics D9, D2, and D11.
Clinical validation and release control (D9) involving escalation requiring human in the loop review increases cost variability due to queued workload depending on case demands of the human, backlog of reviews, and overall case load. Sudden case review demand spikes require excess capacity to absorb that demand, leading to increased costs variation. Training our own model to utilize as our **model origin** (D2) adds retraining cycles, infrastructure costs, and greater operational variation over time, driven by data curation and development efforts. Requiring the **model specification** (D11) to be a more powerful and multimodal model increases cost uncertainty requiring larger context windows and heavier compute burdens.
- **Summary Insight:** The above decisions drive the majority of the dispersion in the simulated tradespace. Compliance-heavy concepts, favored in OS13 (e.g., C7, C10), exhibit greater cost variance due to escalation and high-tier model usage. These architectures lose relative advantage under uncertainty because added assurance increases cost variability without proportionally stabilizing performance.

Acceptability and Mitigation

We defined acceptable uncertainty as:

- Performance: 95% confidence interval within $\pm 10\%$ of the mean
- Cost: 95% confidence interval within $\pm 40\%$ of the mean

Architectures relying on intensive human review and multimodal models exceed acceptable cost variance thresholds, particularly in worst-case (upper tail) outcomes. To reduce these uncertainties, we propose:

1. Targeted validation pilot to empirically measure escalation rates and review throughput.
2. Set-based configuration development to compare single-agent and small-group reliability before scaling.
3. Progressive release gating (shadow mode before full escalation).
4. Prototype data harmonization study prior to committing to multi-source integration.

These actions directly reduce epistemic uncertainty rather than merely modeling it.

Impact on Preferred Architecture and Next Steps

Under uncertainty, C1 ("Maximalist") and C6 ("Prioritize Customizability") appear more robust than compliance-heavy alternatives. However, overlapping 95% confidence regions indicate that it is statistically difficult to declare one architecture superior now.

Uncertainty, therefore, shifts our prioritization:

1. First, resolve uncertainty in D9 (validation policy) and D6 (data coverage).
2. Delay irreversible investment in high-cost multimodal scaling.
3. Continue evaluating C1 and C6 as leading candidates pending validation results.

In summary, uncertainty does not eliminate high-assurance architectures, but it reduces confidence in their relative advantage. Our next phase should focus on reducing the structural uncertainties that most affect valued outcomes before committing to a final architectural decision.

Q4: Reflection

Architectural Reconsideration: The variance in our Monte Carlo simulation solidifies the decision to use our second decomposition from OS13's Q4. The uncertainty demands that we separate the volatile clinical librarian agentic AI roles into modular components decoupling them from the more stable supporting components. This supports concept C1 ("Maximalist"), which shows good cost to performance despite instability in AI providers as they rapidly develop. This architecture could be challenged if agentic AI becomes so proficient as to subsume all of Electronic Medical Record (EMR) handling, but even then, the industry will still need an EMR interface. In this scenario, Sanguine would fulfil not just a clinical librarian role, but also authentication and access, intent, data retrieval, interpretation, review, or auditing and publication in a single monolithic stage. The approach we have chosen allows for shifting to a different model or provider if shifts in performance warrant it.

Non-architectural uncertainty: We focus on data governance, user adoption, competition, and AI turbulence. Existing regulatory frameworks, such as HIPAA (US) and GDPR (EU), cover privacy, but don't fully address agentic AI. There is significant discussion around this topic in the literature. The paper "*Ethical and legal considerations in healthcare AI: innovation and policy for safe and fair use*", highlights "the importance of multi-disciplinary collaboration between technologists, healthcare providers, legal experts and policymakers to create adaptive, globally harmonized frameworks."^[2] It is not clear how regulations will cover AI or how it'll impact our architecture. This uncertainty is a risk, especially since Mayo Clinic Labs' tests have global customers. To mitigate this risk, we should track this area as project development continues.

We consider user adoption and the value Sanguine creates for external (non-Mayo Clinic) physicians. There are two sub-concerns here: whether they find the access to curated information insightful, and whether they integrate it into their workflows. Regarding the former, results could be too expansive to be actionable, or too high-level to be meaningful. Existing literature highlights how an overwhelming Clinical Diagnostic Support System (CDSS) can cause alert fatigue, making physicians ignore the results^[1]. Tuning of the system to combat this is discussed in the next section of development approaches, but further validates our preference to decision C1. Third is the competitive landscape, which must be considered since the primary role of our architecture interfaces is externally facing patients and their physicians. The value of Sanguine could be diminished if an embedded alternative medical diagnostic assistant within their existing workflow is already utilized by physicians.

Finally, the fourth and largest area of non-architectural uncertainty rests with the turbulence of the AI industry. This was a topic of conversation with our sponsor, Dr. Chris Garcia, who mentioned that Mayo Clinic is already reviewing a multi-cloud architecture for internal review to lower uncertainty. Market dominance through subsidized costs to gain market share could lead to unrealistic long-term cost estimates. An example is how Uber and other rideshare companies were able to usurp existing taxi services through low-cost fares^[3], but consumers ended up with a higher-cost service once that dominance was achieved. We may see the same trend followed as the competitive landscape shifts and filtering of existing AI providers occurs, shifting our cost estimates. Alternatively, there was a significant shift felt within the LLM industry when Deepseek provided a lower-cost alternative architecture through distillation^[4] that achieved remarkable results. There could be similar or larger architectural breakthroughs in model design and application that could massively reduce cost. This cost variance is something modeled in our Monte Carlo simulation, and there seems to be some stability in understanding of these costs as they are today, but massive uncertainty in the long term as Sanguine manifests. All of these AI challenges also discount the global supply chain that is required to support long-term agentic AI use, the hardware to support their computation, the datacenters they reside in, and the energy to power them. The long tail risk of disruption due to supply chain turmoil or larger geopolitical uncertainty should be considered.

Analyses and Development Approaches: Waterfall often requires that requirements are locked in fairly early in the development cycle, while a spiral or agile methodology allows adapting to changing environments. Agile practices seem to align more closely to the requirements of the project, where there could be some delay of key decisions. Agile is more complementary to the turbulent technology development of agentic AI. It would also benefit the team to use continuous development and model what external physicians actually have the capacity to absorb. Deploying disparate approaches in an A/B testing model coupled with evaluation allows us to review which approach yields a higher NPS_[A] or PAM_[B] score, key metrics for determining the quality of external physician endpoint value. This strongly aligns with our beachhead strategy, where we provide a meaningful clinical librarian role to end users and then begin to find further areas of expansion of the Sanguine product after establishment. It could also allow for future platform formation if deemed reasonable and valuable after the initial architecture is delivered, but as discussed in OS13, it is still out of scope for the current project. Finally, in our program management, we may want to consider some margins in design to support fluctuations of new models or shifts in pricing, given the uncertainty evident in the deployment of our system. Collectively, these developmental approaches should align with the needs of Sanguine and the collective outcome goals between our team and our sponsor.

Appendix: Code link

Our system model is publicly visible in Colab at

<https://colab.research.google.com/drive/1oDWIRx6uxCVfDNnWWguxeaOVITwkunYh>. We are not expecting a TA to read it, but we're providing it in case you're curious.

Appendix: AI Disclaimer

AI was used to aid generating code implementing the PDFs and the Monte Carlo simulation. However, AI was *not* used to select the PDFs or the parameters quantifying the uncertainty associated with each architectural decision. Google's Gemini was used to review drafts of this assignment and act as the role of a MITsdm TA providing feedback on sections with strong arguments and areas requiring improvement. All work in response to this feedback is the author's own.

Appendix: Key Metrics From OS9+10

- A. **Net Promoter Score (NPS):** (-100 - 100) Willingness to recommend this system. We want users to be delighted by our system.
- B. **Patient Activation Measurement (PAM) Score:** (0-100) Quantifies how well prepared a patient is to manage their health. High PAM scores are correlated with better health outcomes, so an increase in PAM would imply improved patient outcomes.

Appendix: Works Cited

- [1] O. O. Olakotan and M. Mohd. Yusof, "Evaluating the alert appropriateness of clinical decision support systems in supporting clinical workflow," *Journal of Biomedical Informatics*, vol. 106, no. 1, p. 103453, Jun. 2020, doi: <https://doi.org/10.1016/j.jbi.2020.103453>.
- [2] Pham T. 2025 Ethical and legal considerations in healthcare AI: innovation and policy for safe and fair use. R. Soc. Open Sci. 12:241873. <https://doi.org/10.1098/rsos.241873>
- [3] K. Roose, "Farewell, Millennial Lifestyle Subsidy," *The New York Times*, Jun. 08, 2021. Available: <https://www.nytimes.com/2021/06/08/technology/farewell-millennial-lifestyle-subsidy.html>
- [4] D. Seetharaman and F. Arámburo, "OpenAI says China's DeepSeek trained its AI by distilling US models, memo shows," *Reuters*, Feb. 12, 2026. Available: <https://www.reuters.com/world/china/openai-accuses-deepseek-distilling-us-models-gain-advantage-bloomberg-news-2026-02-12/>